

OS ORANGE PROBLEM

NAME : CHINMAYI HEBBAR AM

SRN: PES2UG24CS134

SEMESTER : 4C

GITHUB LINK: <https://github.com/useramc/PES2UG24CS134-pes-vcs.git>

SCREENSHOTS

Phase 1A

```
chinmayi_hebbar@Chinmayi- [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Chinmayi Hebbar A M\Documents\sem 4\OS\OS-orange2\PES2UG24CS134-pes-vcs>wsl
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES
2UG24CS134-pes-vcs$ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES
2UG24CS134-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES
2UG24CS134-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES
2UG24CS134-pes-vcs$ |
```

1b

```
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES
2UG24CS134-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES
2UG24CS134-pes-vcs$
```

2A

```
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4
/OS/OS-orange2/PES2UG24CS134-pes-vcs$ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objec
ts.o test_tree.o
rm -rf .pes
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4
/OS/OS-orange2/PES2UG24CS134-pes-vcs$ make test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4
/OS/OS-orange2/PES2UG24CS134-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4
/OS/OS-orange2/PES2UG24CS134-pes-vcs$
```

2B

```
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4
/OS/OS-orange2/PES2UG24CS134-pes-vcs$ xxd .pes/objects/d5/8213f5dbe0629b5c2fa28e5c
7d4213ea09227ed0221bbe9db5e5c4b9aafc12
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                ES-VCS!.
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4
/OS/OS-orange2/PES2UG24CS134-pes-vcs$
```

```
head -20: Command not found
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pe
s-vcs$ xxd .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12 | head -20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                ES-VCS!.
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pe
s-vcs$
```

3A

```
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$ echo "hello" > file1.txt
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$ echo "world" > file2.txt
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$ ./pes add file1.txt file
2.txt
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$ ./pes status
Staged changes:
  staged:    file1.txt
  staged:    file2.txt

Unstaged changes:
(nothing to show)

Untracked files:
  untracked: .gitignore
  untracked: commit.c
  untracked: commit.h
  untracked: index.c
  untracked: index.h
  untracked: Makefile
  untracked: object.c
  untracked: pes.c
  untracked: pes.h
  untracked: README.md
  untracked: test_objects
  untracked: test_objects.c
  untracked: test_sequence.sh
  untracked: test_tree
  untracked: test_tree.c
  untracked: tree.c
  untracked: tree.h
```

3B

```
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24
CS134-pes-vcs$ cat .pes/index
100755 2cf8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 1776697275 6 file1.txt
100755 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776697279 6 file2.txt
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
```

```
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24
CS134-pes-vcs$ cat .pes/index
100755 2cf8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 1776697275 6 file1.txt
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
```

4

```
bash: ./: is a directory
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
./pes init
Initialized empty PES repository in .pes/
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
echo "Hello" > hello.txt
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
./pes add hello.txt
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
./pes commit -m "Initial commit"
Committed: 5832f126fddb... Initial commit
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24
CS134-pes-vcs$ |
```

```
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24
CS134-pes-vcs$ echo "World" >> hello.txt
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
./pes add hello.txt
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
./pes commit -m "Add world"
Committed: c7cdef0f8438... Add world
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
```

```
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
echo "Goodbye" > bye.txt
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
./pes add bye.txt
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
./pes commit -m "Add farewell"
Committed: 6ebcc49bab7d... Add farewell
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
```

4A

```
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
./pes log
commit 6ebcc49bab7dd199fbb1e8472a097be7a9aa880f7f0e8e424ea6b09e8848d1b9
Author: PES User <pes@localhost>
Date: 1776698481

    Add farewell

commit c7cdef0f8438cdf25d4361b84efelbacdc2fb4a5edf908f611a3323e369266ae
Author: PES User <pes@localhost>
Date: 1776698442

    Add worldell

commit 5832f126fddbaa13e302d98a924aee5bf95fc81d40323f0ce9f1e5bb9c99fb67
Author: PES User <pes@localhost>
Date: 1776698392

    Initial committial commiQ

chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
```

4B

```
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/55/8c1061e21bccbeb23c527d8dcef5c8646cabfb6091aae3ae7ee3320c9ff7b7
.pes/objects/58/32f126fddbaa13e302d98a924aee5bf95fc81d40323f0ce9f1e5bb9c99fb67
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a523d5f4aa8007b5c39
.pes/objects/6e/bcc49bab7dd199fbb1e8472a097be7a9aa880f7f0e8e424ea6b09e8848d1b9
.pes/objects/c7/cdef0f8438cdf25d4361b84efelbacdc2fb4a5edf908f611a3323e369266ae
.pes/objects/cc/e49eb2943f54e8135f291862bdf218064409ba05a7ee7d5bec00ddc22adddf
.pes/objects/dd/82340b5e403e2dc18bfca050e6699935849710b2b5d28ab8bbffbb72d32627
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
```

4C

```
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
cat .pes/refs/heads/main
6ebcc49bab7dd199fbb1e8472a097be7a9aa880f7f0e8e424ea6b09e8848d1b9
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
cat .pes/HEAD
ref: refs/heads/main
chinmayi_hebbar@Chinmayi-Hebbar09:/mnt/c/Users/Chinmayi Hebbar A M/Documents/sem 4/OS/OS-orange2/PES2UG24CS134-pes-vcs$
```

Analysis Questions

Q5.1

Implementing `git checkout <branch>` means updating `.git/HEAD` to point to `refs/heads/<branch>` and reading that branch's commit to restore its tree. The index must be replaced with the target tree's entries, and the working directory files must be rewritten to match that snapshot. This is complex because files may need to be created, deleted, or modified while preserving untracked files and handling conflicts safely.

Q5.2

To detect a dirty working directory, compare each tracked file's metadata (mtime and size) in the index with the current filesystem state using `stat()`. If a file differs and its content hash (computed again if needed) does not match the stored hash, it is modified. Then compare this against the target branch's tree; if both differ, refuse checkout due to conflict.

Q5.3

In a detached HEAD state, new commits are created but not referenced by any branch, so they can become unreachable. These commits remain in the object store temporarily but may be lost during garbage collection. A user can recover them by creating a new branch pointing to the commit hash before they are deleted.

Q6.1

Start from all branch heads and traverse commits, trees, and blobs using a graph walk, marking each visited object as reachable. Use a hash set for efficient tracking of visited object IDs. After traversal, delete any unmarked objects. For 100,000 commits and 50 branches, traversal typically touches all reachable commits plus associated trees and blobs, potentially hundreds of thousands of objects.

Q6.2

Running garbage collection during a commit can cause a race where GC deletes an object that a commit is about to reference but hasn't yet linked. For example, a blob written but not yet referenced by a commit could be seen as unreachable and removed. Real Git avoids this using object locking, temporary references, and running GC only when no write operations are in progress.